Continuous and Contactless Blood Pressure Monitoring Using a Hybrid Neural Network Model Based on Facial Video Analysis

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ABSTRACT

Blood pressure (BP) is a critical vital sign for evaluating cardiovascular health, and hypertension, defined as persistently elevated BP, affects nearly half of U.S. adults and contributes to over 10 million deaths worldwide each year. Despite its prevalence and impact, hypertension remains underdiagnosed and poorly managed due to the lack of tools capable of continuous and real-time BP monitoring. Conventional cuff-based devices are limited to intermittent, manual measurements, failing to capture BP fluctuations caused by activity, stress, or medication. To address this unmet need, we propose a hybrid neural network (HNN) model capable of accurate, continuous, and contactless BP inference from facial video input. The proposed system integrates facial detection, adaptive cheek region-of-interest (ROI) extraction, and AI-based BP estimation into a unified framework. This model can derive one BP reading per second from either live or recorded facial videos, providing acceptable accuracy without physical contact. The contactless and real-time nature of this approach enhances usability and accessibility, offering significant potential for remote health monitoring, chronic disease management, and elder care applications. The HNN model has been validated on a large dataset comprising high-resolution facial videos and BP recordings, demonstrating its promise as a transformative tool for digital health and personalized hypertension management.

Keywords: Vital sign, Facial video analysis, Hybrid neural network (HNN), Real-time blood pressure monitoring, Digital health, Contactless vital sign measurement.

1. INTRODUCTION

Vital signs, including body temperature, heart rate (HR), respiration rate (RR), blood pressure (BP), and blood oxygen saturation (SpO_2), are fundamental indicators of physiological status and are critical for detecting clinical deterioration and guiding timely intervention. Continuous monitoring of these signals is particularly important in acute care settings, where patients are at elevated risk and require close surveillance. However, conventional monitoring systems are predominantly contact-based and often provide only intermittent measurements. These devices can be cumbersome, intrusive, and inconvenient for long-term use, thereby limiting patient compliance and the ability to capture dynamic physiological changes. In contrast, non-contact, video-based approaches offer a flexible and unobtrusive alternative, with the potential to enable continuous, scalable, and data-rich monitoring.

Among vital signs, BP is especially important for assessing cardiovascular health. Hypertension affects nearly half of adults in the United States and remains a leading contributor to heart disease, stroke, kidney failure, and other chronic conditions worldwide. Despite its significant public health impact, hypertension is frequently underdiagnosed and poorly controlled, partly due to the lack of continuous monitoring tools. BP can fluctuate substantially due to physical activity, stress, medication, and underlying health conditions, making real-time tracking essential for accurate diagnosis and effective management. However, current cuff-based BP devices are limited to sporadic measurements and require active user participation, resulting in poor compliance and an inability to detect transient yet clinically significant BP variations.

Remote photoplethysmography (rPPG) has emerged as a promising non-contact technique for vital sign monitoring by extracting subtle physiological signals from facial videos using standard cameras. Prior studies have demonstrated the

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feasibility of rPPG for estimating HR, RR [1], BP [2], and SpO_2 [3], with facial regions providing reliable signal sources due to their visibility and vascular structure. Nevertheless, rPPG-based methods face significant challenges, including weak signal strength, susceptibility to motion artifacts (e.g., head movement and facial expressions), and variations in ambient illumination and camera noise. Additionally, the scarcity of large, high-quality annotated datasets further complicates the development of robust models, particularly for deep learning-based approaches.

Traditional signal processing methods such as independent component analysis (ICA), principal component analysis (PCA) [4,5], and wavelet analysis [1] have been widely applied to rPPG signal extraction but often struggle under real-world conditions with motion and lighting variability. To address these limitations, recent research has increasingly focused on deep learning techniques that directly learn spatiotemporal representations from video data. For example, attention-based models have been proposed to identify informative facial regions and improve HR and RR estimation [6–9], while multi-level convolutional architectures further enhance feature extraction and prediction accuracy [10]. In the context of BP estimation, methods leveraging neural networks, generative adversarial networks (GANs), and ResNet-based architectures have shown promising results by modeling complex relationships between rPPG signals and BP values [11–13].

In addition, approaches based on skin reflection models [15] and advanced deep neural networks [14,16–18] have further improved the robustness of remote physiological measurement by mitigating motion artifacts and enhancing signal quality. However, these methods typically require large volumes of synchronized video and physiological data for supervised training, which are difficult and costly to collect. Recent efforts have explored self-supervised learning strategies to reduce reliance on labeled data by leveraging intrinsic temporal and frequency properties of rPPG signals [19]. Despite these advances, developing accurate, robust, and data-efficient solutions for continuous, non-contact BP monitoring remains an open and important research challenge.

1.1 Related Work

Early studies have explored the use of deep learning and video-based representations to estimate physiological signals from facial videos. For example, Hsu et al. [20] proposed a method for HR estimation by transforming short temporal segments of facial videos into 2D time–frequency representations, which were then used as input feature maps for a convolutional neural network (CNN). Similarly, Omer et al. [21] developed a beat-to-beat BP estimation framework that converts one-dimensional pulse signals into two-dimensional image representations and applies transfer learning using a pre-trained ResNet-101 model. Their approach incorporates signal quality assessment by filtering invalid beats prior to training, achieving mean absolute errors (MAE) of 7.05 mmHg for systolic BP and 5.62 mmHg for diastolic BP on real-world video data.

More recent work has focused on leveraging advanced deep learning architectures to improve estimation accuracy by capturing complex spatiotemporal dynamics. Zheng [22] introduced a framework for directly predicting HR and SpO_2 from facial videos by comparing seven neural network architectures. A hybrid model integrating CNN, convolutional LSTM (convLSTM), and Vision Transformer (ViViT) components achieved the best performance, demonstrating superior capability in extracting temporospatial features and yielding the lowest root mean square errors (RMSEs). In parallel, a growing body of research [23–26] has demonstrated the feasibility of using artificial intelligence (AI) and facial video analysis to estimate multiple vital signs, including BP, by analyzing subtle variations in skin tone and facial features associated with blood flow.

Despite these advances, existing video-based BP estimation methods still face significant limitations. Many approaches lack real-time processing capability or fail to achieve the accuracy required for clinical or consumer applications. In addition, most prior studies follow a two-stage pipeline in which rPPG signals are first extracted from facial videos and then used as inputs to separate models for estimating individual vital signs, typically HR or BP. This separation may limit overall system efficiency and robustness.

To address these gaps, we propose an improved hybrid neural network (HNN) model that integrates facial detection, BP inference, and continuous monitoring into a unified framework. The proposed model combines CNN, convLSTM, and ViViT architectures to directly estimate BP values from facial videos, enabling end-to-end learning of spatiotemporal features. The remainder of this paper is organized as follows: Section 2 describes dataset preprocessing, Section 3 reviews the deep learning models, Section 4 presents experimental results, and Section 5 concludes the paper.

2. DATA PREPROCESSING

To ensure that the proposed Hybrid Neural Network (HNN) model is trained on clean and reliable data, thereby enhancing predictive performance, we implemented a series of rigorous preprocessing steps, including data normalization, facial detection, cheek region extraction, and data standardization. The face detection and cheek region-of-interest (ROI) extraction modules serve as quality control mechanisms to filter out low-quality inputs. If either face detection or cheek ROI extraction fails, the corresponding process is terminated, indicating inadequate video quality. Such data (video frames) are discarded during the training phase and skipped during the testing phase to maintain robustness and accuracy.

2.1 Image normalization and face detection

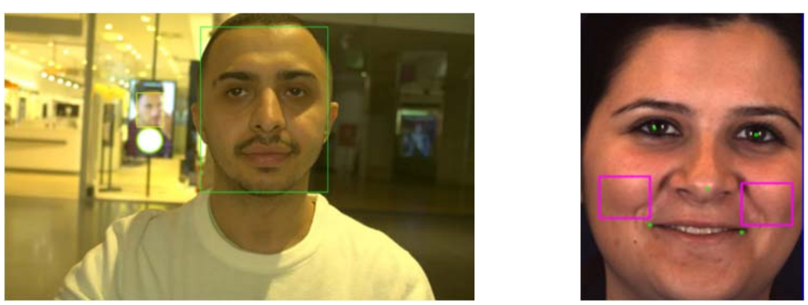
Image normalization (intensity scaling/stretching) is defined as

$$\mathbf{I}_{\mathrm{N}} = (\mathbf{I}_0 - I_{\mathrm{Min}})\frac{L_{\mathrm{Max}} - L_{\mathrm{Min}}}{I_{\mathrm{Max}} - I_{\mathrm{Min}}} + L_{\mathrm{Min}} \tag{1}$$

where $\mathbf{I}_{\mathrm{N}}$ is the normalized image, $\mathbf{I}_0$ is the original image; I_{Min} and I_{Max} are the minimum and maximum pixel value in $\mathbf{I}_0$, while L_{Min} and L_{Max} are the desired minimum and maximum pixel value in $\mathbf{I}_{\mathrm{N}}$. For example, we may select $L_{\mathrm{Min}} = 0$ and $L_{\mathrm{Max}} = 255$.

Face detection aims to localize facial regions in images and serves as a critical preprocessing step for subsequent analysis. A wide range of approaches have been developed, including knowledge-based methods, feature-invariant techniques, template matching, and appearance-based models, as comprehensively reviewed in [27]. Many traditional methods leverage skin-color information to reduce the search space and improve efficiency. Among these, the Viola–Jones (VJ) algorithm [28] is one of the most influential approaches, employing integral images [29,30] for efficient feature computation, an AdaBoost-based learning framework [31] for feature selection and classification, and a cascade of classifiers to progressively filter out non-face regions. This cascade structure enables rapid detection by discarding unlikely regions early while focusing computation on promising candidates.

More recently, deep learning-based methods have significantly advanced face detection performance. The Multi-task Cascaded Convolutional Neural Network (MTCNN) [32–34] is a widely adopted framework that integrates face detection and alignment through a three-stage cascade of convolutional networks: Proposal Network (P-Net), Refine Network (R-Net), and Output Network (O-Net). Following a coarse-to-fine strategy, P-Net generates candidate face regions across multiple scales, R-Net refines these proposals, and O-Net produces final bounding boxes (Fig. 1a) along with facial landmarks (eyes, nose and mouth shown as green dots in Fig. 1b). By jointly optimizing face classification, bounding box regression, and landmark localization, MTCNN achieves improved accuracy and efficiency. Its use of image pyramids for scale invariance and non-maximum suppression (NMS) to eliminate redundant detections further enhances performance. Although newer detectors may outperform MTCNN in some benchmarks, it remains widely used due to its balance of accuracy, speed, and simplicity, particularly in real-time and preprocessing applications. In our study, we utilized a pre-trained MTCNN model from the OpenCV package for facial detection.



(a) (b)

Fig. 1 Illustration of (a) MTCNN face detection (ROI: 288×256), and (b) cheek area extraction (ROI: 36×96).

2.2 Cheek ROI extraction and data standardization

Cheek region extraction is performed based on detected facial landmarks after face detection and resizing to 288×256 pixels (Fig. 1a). Specifically, the cheek regions of interest (ROIs) are defined using the positions of the two mouth corners obtained from the MTCNN model, with fixed-size patches (36×48) extracted from each side of the face (Fig. 1b). The left and right cheek regions are then concatenated to form a single image (36×96) for subsequent processing. Focusing on cheek ROIs helps reduce interference from occlusions and variations such as hair, glasses, facial hair, and expressions, thereby improving signal quality for BP estimation.

We furhter apply general standardization all cheek-ROI images.

$$\mathbf{I}'_{\mathrm{N}} = \mathbf{I}_{\mathrm{N}} - \mathbf{I}_{\mathrm{M}} \, , \tag{2}$$

$$\mathbf{I}_{\mathrm{S}} = (\mathbf{I}'_{\mathrm{N}} - \mu)/\sigma \tag{3}$$

where $\mathbf{I}_{\mathrm{N}}$ is the normalized facial image, $\mathbf{I}_{\mathrm{M}}$ is the mean image of all faces in the dataset, and $\mathbf{I}'_{\mathrm{N}}$ is their difference image. $\mathbf{I}_{\mathrm{S}}$ is the standardized image; μ and σ denote the mean and standard deviation of the $\mathbf{I}'_{\mathrm{N}}$ image, respectively.

All facial frames are processed using image normaliztion and face detection by the MTCNN model, followed by cheek ROI extraction and standardization. The resulting cheek-region video sequences are segmented into 36-frame clips to serve as input samples, while corresponding BP measurements are used as ground-truth outputs, including both systolic and diastolic BP. As these outputs are continuous variables, the proposed HNN model is formulated as a regression framework to directly predict BP values from the input video clips.

3. HYBRID NEURAL NETWORKS

The proposed HNN model integrates CNN, convLSTM, and ViViT architectures. This section briefly reviews each component before introducing the HNN model design.

3.1 Convolutional neural network

Convolutional neural networks (CNNs), inspired by the visual cortex, have become fundamental in computer vision. Their prominence grew after AlexNet [29], an 8-layer CNN, significantly improved ImageNet performance, reducing classification error from 25.8% to 16.4%. Since then, CNNs have driven major advances in deep learning applications. AlexNet [35] contains five convolutional and three fully connected layers, achieving a top-5 error rate of 15.3% in ILSVRC 2012 [36] with 63M parameters. VGG-19 [37,38] increased depth using 3×3 filters, achieving a 7.3% error rate and ranking highly in ILSVRC 2014, though with 143.7M parameters. ResNet models address training difficulties in deep networks using residual connections. ResNet-101 [39], with over 100 layers, enables stable training by learning residual mappings, achieving top performance in ILSVRC 2015.

3.2 Video Vision Transformer

Transformers, originally developed for NLP tasks such as BERT [40], model relationships using attention mechanisms. Direct application to images is computationally expensive due to pixel-level interactions. Vision Transformers (ViTs) address this by dividing images into smaller patches (e.g., 16×16), embedding them into sequences with positional encoding, and processing them through transformer layers.

The Video Vision Transformer (ViViT) extends ViT by incorporating temporal information using video tube embeddings (e.g., frame × height × width). Self-attention generates Query, Key, and Value vectors, while multi-head attention enables the model to capture diverse feature relationships across spatial and temporal dimensions.

3.3 Recurrent Neural Network

Recurrent neural networks (RNNs) are designed to model sequential dependencies by maintaining hidden states across time. They have been widely applied in tasks involving temporal data [41], though standard RNNs struggle with long-term dependencies. Long short-term memory (LSTM) networks address this limitation through gated mechanisms that regulate information flow, enabling effective long-term memory retention.

In applications such as vital sign estimation, both temporal dynamics (e.g., physiological signal variation) and spatial features are important. Convolutional LSTM (ConvLSTM) combines convolutional operations with recurrent structures, allowing simultaneous modeling of spatial and temporal patterns, making it well-suited for video-based physiological signal analysis.

3.4 Hybrid models

Given facial video clips as Input (36×36×96×3) and two continuous Outputs, systolic BP (SBP) and diastolic BP (DBP), the proposed neural network follows a standard structure consisting of Input, Body, Head, and Output blocks. As shown in Table 1, the time_distr operator applies identical convolutional operations across all 36 frames. The Head block contains two fully connected layers, with separate branches for SBP and DBP, whose outputs are concatenated into a final prediction vector. The Body component includes two modules: ViViT (Mdl_a) and ConvLSTM (Mdl_b), which can be used independently (by omitting the feature Concat operator) or combined via feature concatenation to form a hybrid model (Hybrid_Model_ab).

The ViViT Body block (Mdl_a) comprises a Tubelet embedding stage followed by spatial and temporal attention mechanisms. The embedding stage uses two 3D convolutional layers to generate patch representations. A shallow spatial attention module (2 layers) incorporates positional encoding, multi-head attention (4 heads), and a feed-forward network (FFN), followed by a deeper temporal attention module (24 layers) with positional encoding, multi-head attention (16 heads), and FFN. With a single-frame tubelet for preserving temporal resolution, Mdl_a produces 1,728 embedding vectors (512 dimensions each) from 36 frames and 48 spatial patches.

The ConvLSTM Body block (Mdl_b) consists of three ConvLSTM layers with specified filters, kernel sizes, and strides. Spatial features are flattened to preserve both spatial and temporal resolution in the feature space, albeit with increased dimensionality and parameter size.

Table 2 further introduces an extended hybrid model (Hybrid_Model_abc), which incorporates an additional ResCNN block (Mdl_c). This module includes three time-distributed convolutional layers, each with a residual block, followed by flattening of spatial patches. The combination of ViViT, ConvLSTM, and ResCNN enables complementary modeling of spatial, temporal, and hierarchical features for improved BP estimation.

Table 1. Hybrid Model (HMdl_ab) architecture — combining 2 NN models: ViViT and ConvLSTM. Normalization and Dropout layers are omitted. The batch size (B) is omitted in the "Output Shape" column except for one row ("Positional_Encoder for Space") in the ViViT model.

Hybrid_Model_ab: ViViT ⊕ ConvLSTM, 1620.8M Paras			
Layer (Type)	***Output Shape***	***Layer (Type)***	***Output Shape***
Frame_Input	(36, 36, 96, 3)		
***ViViT** (Mdl_a)*:		***ConvLSTM** (Mdl_b)*:	
Tubelet_Embedding:		(filters, kernel_size, strides) =	
(Conv3D → 36 × 12 × 24 patches)	(36, 48, 512)	(128, (3,4), (3,4))	(36, 12, 24, 128)
(Conv3D → 36 × 4 × 12 patches)		(256, (2,2), (3,2))	(36, 4, 12, 256)
Add (Positional_Encoder for Space)	(B*36, 48, 512)	(512, (1,2), (1,2))	(36, 4, 6, 512)
2 × Attn_Space_FF:			
MultiHeadAttention:			
(heads = 4, key_dim = 128)			
Feed_Forward_Net:			
(512 → 2048→512)			
Add (Attn, FF)	(48, 512)		
Add (Positional_Encoder for Time)	(36*48, 512)		
24 × Attn_Time_Space_FF:			
MultiHeadAttention:			
(heads = 16, key_dim = 32)			
Feed_Forward_Net:			
(512 → 2048→512)			
Add (Attn, FF)	(1728, 512)		
Flatten	(884736)	Flatten	(442368)
Concat (ViViT, convLSTM)	(1327104)		
2 × ***Head***:			
Dense	(576)		
Dense	(1)		
Concat (Output)	(2)		

Table 2. Hybrid Model (HMdl_abc) architecture — combining 3 NN models: ViViT, ConvLSTM and ResCNN. Normalization and Dropout layers are omitted. The batch size is omitted in the "Output Shape" column. time_distr = time_distributed.

Hybrid_Model_abc: ViViT ⊕ ConvLSTM ⊕ ResCNN, 2131.2M Paras			
Layer (Type)	***Output Shape***	***Layer (Type)***	***Output Shape***
Frame_Input	(36, 36, 96, 3)		
***ViViT** (Mdl_a):* (from Hybrid_Model_1) …	***ConvLSTM** (Mdl_b):* (from Hybrid_Model_1) …	***ResCNN** (Mdl_c)*: time_distr (Conv2D): (filters, kernel_size, strides) = (128, (5,5), (3,4)) (256, (3,3), (3,2)) (512, (3,3), (1,2))	 (36, 12, 24, 128) (36, 4, 12, 256) (36, 4, 6, 512)
Flatten (884736)	Flatten (442368)	Flatten	(442368)
Concat (ViViT, convLSTM, ResCNN)	(1769472)		
2 × ***Head***:			
Dense	(576)		
Dense	(1)		
Concat (Output)	(2)		

3.5 Seven Neural Network (NN) models

Using three base models, ViViT (Mdl_a), ConvLSTM (Mdl_b), and ResCNN (Mdl_c), we construct four hybrid variants: Mdl_ab, Mdl_ac, Mdl_bc, and Mdl_abc (see Tables 1 and 2). Together with the three individual models, a total of seven neural network architectures are evaluated in our experiments, with parameter sizes summarized in Table 3.

In these models, CNN-based components process individual frames, while ConvLSTM and ViViT capture temporal dependencies across frame sequences. Given the dynamic nature of BP, the proposed HNN framework emphasizes high temporal resolution during feature extraction. By integrating temporally informative features with spatial representations, the model effectively captures BP variability and improves prediction accuracy..

4. EXPERIMENTAL RESULTS

4.1 Vital Video Dataset

Data-driven AI models depend critically on both data quantity and quality. In this study, we utilized a large-scale dataset, the Vital Video Dataset (VVD) [22, 42], comprising 891 participants (459 female, 432 male). For each participant, two 30-second compressed facial videos were recorded along with synchronized PPG waveforms and a single BP measurement (by a cuff-style BP monitor). Additionally, heart rate (HR) and blood oxygen saturation (SpO_2) values were obtained using a pulse oximeter. Importantly, BP was measured during video acquisition to ensure temporal consistency, as measurements taken before or after recording may not accurately reflect physiological conditions.

Facial videos were captured at 30 frames per second (FPS) with a resolution of 1920×1200. Since a typical cardiac cycle is less than one second, 36 consecutive frames (1.2 seconds) were used as input sequences for the neural network models. For each subject, four non-overlapping 36-frame clips were randomly extracted from the two recorded videos. During dataset partitioning (e.g., training and testing splits), all clips from the same subject were assigned exclusively to one subset (either train set or test set) to prevent data leakage.

All frames underwent face detection and alignment, followed by resizing to 288×256 pixels. Cheek ROI of size 36×96 were then extracted and standardized as described in Section 2. These processed video clips served as model inputs, while systolic and diastolic BP values were used as continuous output variables. These ground-truth BP values were normalized to the range [0,1] during training and rescaled to their original units during inference.

Due to the large dataset size and model complexity, training is computationally intensive (see Table 3). Therefore, instead of 10-fold cross-validation, we adopted a random but fixed split, using 90% of the data (802 subjects) for training and 10% (89 subjects) for testing. All reported results are based on the testing set. As illustrated in Fig. 2, training and validation losses converge after approximately 70 epochs. Training was terminated early if the validation loss did not improve for 14 consecutive epochs or when the maximum of 128 epochs was reached.

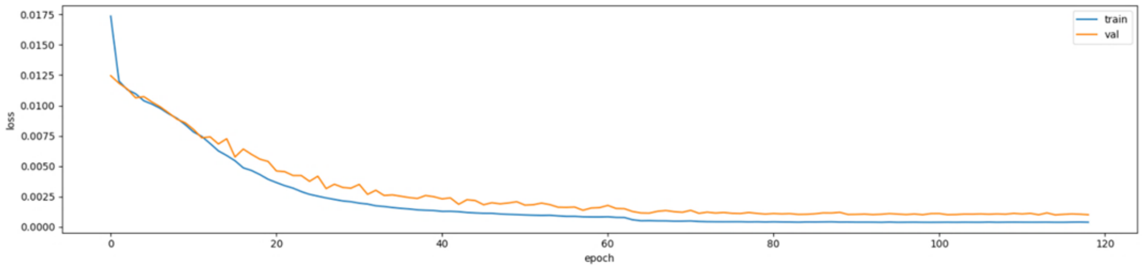


Fig. 2 Train and validation (10% of train dataset) loss over 119 epochs (HMdl_ac, 1620.8 MM, 15.7 hours of training)

4.2 Seven NN model performance

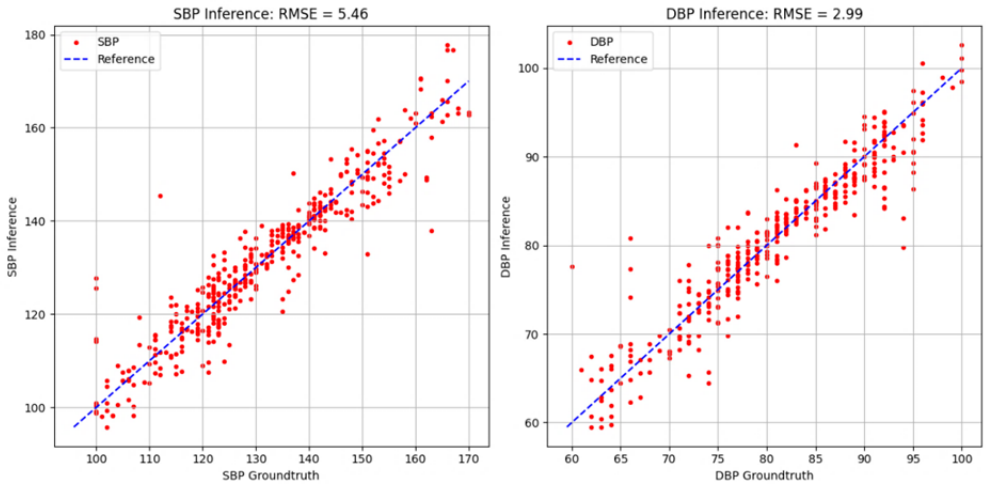
The performance of the neural network models is evaluated using root mean square error (RMSE) and root mean square deviation (RMSD), which quantify the differences between predicted and ground-truth values. Lower RMSE indicates better predictive accuracy. RMSD is influenced by the distribution range of each vital sign; therefore, comparisons of RMSD are meaningful across models for the same variable, but not across different vital signs.

Table 3. The root mean square errors (RMSEs) and the root mean square deviations (RMSDs) of systolic and diastolic blood pressure (SBP, DBP) prediction varying with seven deep learning NN models: Trained on the data samples from 90% subjects and Tested on the samples from 10% subjects. At left column, the model parameters and its output features (to Heads) are presented. At right two columns, the time cost for training and testing (on Nvidia H100 80GB-GPU), total number of training epochs, the minimal validation loss are provided.

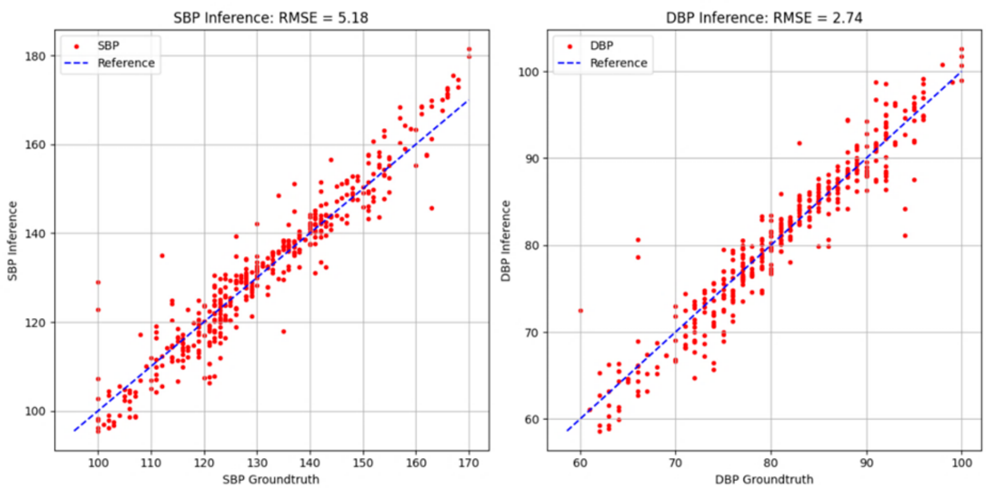
Model \ Blood Pressure	**SBP**	**DBP**	**Train** s/epoch, **#Epochs**	**Test** ms/sample, **val_loss**
Mdl_a, 1110.3M ViViT: 884736	9.11 (18.01)	5.20 (10.43)	455s, 109	91ms, 0.0032
Mdl_b, 599.8M ConvLSTM: 442368	12.11 (14.21)	7.17 (7.66)	181s, 128	66ms, 0.0058
Mdl_c, 599.8M ResCNN: 442368	12.78 (13.93)	7.42 (8.24)	162s, 117	146ms, 0.0065
HMdl_ab, 1620.8M 1327104	6.09 (17.72)	3.62 (9.99)	496s, 97	106ms, 0.0015
HMdl_ac, 1620.8M 1327104	5.18 (20.29)	2.89 (11.15)	475s, 119	250ms, 0.0010
HMdl_bc, 1110.3M 884736	9.76 (15.57)	5.44 (7.84)	240s, 64	122ms, 0.0036
HMdl_abc, 2131.2M 1769472	5.46 (18.66)	3.24 (10.25)	521s, 91	250ms, 0.0012

Table 3 summarizes the performance of seven NN models for predicting systolic BP and diastolic BP. Models based on ConvLSTM (Mdl_b), ResCNN (Mdl_c), and their combination (HMdl_bc) show relatively lower performance, likely due to lack of temporal modeling or resolution. In contrast, the ViViT model (Mdl_a) effectively captures both spatial and temporal features with attention mechanisms applied to frame and spatial patches, achieving the lowest RMSE among the three base models (_a, _b, _c). Hybrid models incorporating ViViT (Mdl_ab, Mdl_ac, Mdl_abc) further improve performance, with HMdl_ac and HMdl_abc yielding the best results. These findings suggest that combining ConvLSTM and ResCNN with ViViT enhances spatial feature extraction while preserving strong temporal modeling capabilities.

To further illustrate prediction performance, scatter plots of predicted versus ground-truth values are shown in Fig. 3. The ground-truth ranges are SBP: [100, 170] and DBP: [60, 100]. In an ideal scenario, all predictions would lie on the diagonal line (blue dashed line), indicating perfect agreement (RMSE = 0). As observed in Fig. 3, most predictions cluster closely around this line, demonstrating strong correlation and high predictive accuracy of the proposed model.



(a)



(b)

Fig. 3 Systolic and diastoplic BP predictions from top 2 best models: prediction (in red dots) over ground truth. The plots illustrate the correlation between the predictions (from (a) HMdl_abc, 2131.2M parameters; (b) HMdl_ac, 1620.8 M parameters) and the ground truth. The range of ground truth: Range(SBP) = [100,170], Range(DBP) = [60,100]. In an ideal case, all predicted values (red dots) would lie exactly on the blue dashed line, indicating perfect agreement with the ground truth (i.e., RMSE = 0). As shown in the plots above, the majority of red dots closely around the blue line, indicating the excellent predictive performance of our trained models.

4.3 Time costs of seven NN models

All models were implemented using TensorFlow 2.19.1 and executed in JupyterLab (Version 4.3.5) on a high-performance workstation equipped with an Intel Xeon Ice Lake Gold 5317 12-core CPU (3.0 GHz), 512 GB RAM, 12 TB SSD storage, and running Ubuntu 22.04. The system also includes an NVIDIA H100 GPU with 80 GB onboard memory and 14,592 CUDA cores. All training and testing processes were performed using the H100 GPU.

The computational costs of the seven NN models are summarized in Table 3 (right two columns). Among them, the hybrid model HMdl_abc requires the longest training and testing time due to its architectural complexity. Despite this, the inference time is approximately 250 milliseconds per sample (36 frames), enabling the system to process about four samples per second, which is sufficient for real-time BP monitoring applications.

5. SUMMARY AND DISCUSSION

The proposed hybrid neural network (HNN) integrates convolutional neural networks (CNN), convolutional long short-term memory (ConvLSTM), and Video Vision Transformers (ViViT) to estimate continuous blood pressure (BP) from facial videos. This approach addresses the need for non-invasive, continuous, and contactless BP monitoring, particularly in elderly care, hypertension management, and remote patient monitoring programs. The model achieves a temporal resolution of one BP reading per second, with RMSEs of ≤ 5.2 mmHg for systolic BP and ≤ 2.9 mmHg for diastolic BP.

By enabling early detection of abnormal BP patterns and supporting timely intervention, this technology has the potential to improve hypertension diagnosis and management. Better BP control is associated with reduced risks of cardiovascular diseases, including stroke, heart failure, atrial fibrillation, chronic kidney disease, and mortality. Furthermore, improved BP management can decrease emergency visits and hospitalizations, lowering healthcare costs. Integration with telehealth systems and chronic disease management platforms can further extend its impact, promoting accessible and personalized healthcare for underserved populations.

Facial videos can be captured using common devices such as smartphones, laptops, or webcams, eliminating the need for specialized medical equipment. This provides a cost-effective and convenient solution, especially for elderly populations and scenarios involving infectious diseases such as COVID-19. With larger datasets, the accuracy and robustness of vital sign prediction can be further improved. In addition, inference latency can be reduced with advanced GPU clusters, while higher temporal resolution may be achieved using high-frame-rate cameras (e.g., 90FPS) and improved temporal modeling techniques.

6. ACKNOWLEDGMENTS

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